

Numerical Methods for Quants

From PDEs to Monte Carlo: The Desk-Ready Computational Playbook

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Contents

How to Use This Book	2
Module 1: Root Finding & Optimization (The Calibration Engine)	5
Module 2: Interpolation & Curve Construction (Building the Surface)	15
Module 3: Finite Difference Methods (Solving PDEs on a Grid)	23
Module 4: Monte Carlo Methods (The Universal Pricer)	32
Module 5: Trees & Lattice Methods (The Visual Pricer)	41
Module 6: Fourier & Spectral Methods (The Calibration Accelerator)	48
Module 7: Calibration & Model Selection (Fitting Models to Markets)	55
Module 8: Convergence, Stability & Performance (Production Numerics)	62
Module 9: Finite Element Methods (FEM) for Quants	71
Module 10: Machine Learning in Numerical Finance	78
Interview Toolkit: 15 Classic Numerical Methods Questions	87
Mini Exercises: 15 Scenarios with Solutions	91
How This Note Connects to the Other Playbooks	93

1.2 Newton-Raphson: Speed with Risk

Σ Core Formula

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

★ Equation Shortcut: Newton-Raphson Vega Safety

For finding Black-Scholes implied volatility, $f'(x)$ is the option's Vega (v). If the option is deep OTM/ITM, Vega is exponentially small ($v \rightarrow 0$). The Newton update step will divide by a near-zero value, throwing the next guess x_{n+1} to infinity or negative territory. Always clamp the update or fall back to a Bisection step if Vega falls below 10^{-6} .

α Proportional Scaling: Newton-Raphson Error Propagation

The error in step $n + 1$ of Newton-Raphson is directly proportional to the square of the error in step n :

$$e_{n+1} \propto e_n^2$$

This quadratic scaling means that once you are near the root, the number of correct decimal places doubles at every single step (e.g., 2, 4, 8, 16 correct digits).

Proof of Quadratic Convergence: Expand $f(x)$ about the true root x^* using a Taylor series:

$$0 = f(x^*) = f(x_n) + f'(x_n)(x^* - x_n) + \frac{1}{2}f''(x_n)(x^* - x_n)^2 + O((x^* - x_n)^3)$$

Divide by $f'(x_n)$ (assuming $f'(x_n) \neq 0$):

$$0 = \frac{f(x_n)}{f'(x_n)} + x^* - x_n + \frac{1}{2} \frac{f''(x_n)}{f'(x_n)} (x^* - x_n)^2 + O((x^* - x_n)^3)$$

Using the Newton iteration formula, we can substitute $x_{n+1} - x_n = -\frac{f(x_n)}{f'(x_n)}$:

$$x_{n+1} - x^* = \frac{1}{2} \frac{f''(x_n)}{f'(x_n)} (x_n - x^*)^2 + O((x_n - x^*)^3)$$

Let $e_n = x_n - x^*$ be the error at step n :

$$e_{n+1} \approx \frac{f''(x^*)}{2f'(x^*)} e_n^2$$

This proves that the error at step $n + 1$ is proportional to the square of the error at step n (quadratic convergence).

Module 7: Calibration & Model Selection (Fitting Models to Markets)

► Module Overview

What this module is about: Calibration is the process of fitting model parameters to market observables. It is not curve fitting—it is controlled deception. Every calibrated model is wrong; the question is how wrong and where.

Why this matters: A perfectly calibrated model can produce terrible hedges. Calibration fits today's prices; hedging needs tomorrow's dynamics. The gap between these two is model risk.

What confusion this clears:

- Why perfect calibration $\neq$ good model
- How to detect over-fitting vs under-fitting
- When to use local vol vs stochastic vol vs jump models

After this module, you should be able to answer: "Your Heston calibration fits the smile perfectly but your barrier prices are 20% off market. Why?"

7.1 Dupire's Local Volatility Model: Derivation

Dupire's model defines a state-dependent volatility function $\sigma_{\text{loc}}(S, t)$ such that the model matches the entire option price surface exactly.

Let the risk-neutral spot process be $dS_t = (r - q)S_t dt + \sigma_{\text{loc}}(S_t, t)S_t dW_t$. Under the forward Fokker-Planck (Kolmogorov) equation, the transition probability density $p(S, T \mid S_0, 0)$ satisfies:

$$\frac{\partial p}{\partial T} = \frac{1}{2} \frac{\partial^2}{\partial S^2} [\sigma_{\text{loc}}^2(S, T) S^2 p] - \frac{\partial}{\partial S} [(r - q) S p]$$

A European call option price is written as the expectation of the payoff:

$$C(K, T) = e^{-rT} \int_K^\infty (S - K) p(S, T \mid S_0, 0) dS$$

Differentiating C twice with respect to K yields:

$$\frac{\partial C}{\partial K} = -e^{-rT} \int_K^\infty p(S, T) dS, \quad \frac{\partial^2 C}{\partial K^2} = e^{-rT} p(K, T)$$

Differentiating C with respect to T and substituting the Fokker-Planck PDE:

$$\frac{\partial C}{\partial T} = -rC + e^{-rT} \int_K^\infty (S - K) \frac{\partial p}{\partial T} dS$$

Integrating the right-hand side by parts twice yields the Dupire forward equation:

$$\frac{\partial C}{\partial T} = \frac{1}{2} \sigma_{\text{loc}}^2(K, T) K^2 \frac{\partial^2 C}{\partial K^2} - (r - q) K \frac{\partial C}{\partial K} - qC$$

7.3 Stochastic Local Volatility (SLV) Models

Stochastic Local Volatility models combine the exact calibration properties of Local Volatility with the realistic smile dynamics of Stochastic Volatility. The dynamics under the SLV model are:

$$\begin{aligned} dS_t &= (r - q)S_t dt + L(S_t, t) \sqrt{v_t} S_t dW_t^S \\ dv_t &= \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t^v \\ d\langle W^S, W^v \rangle_t &= \rho dt \end{aligned}$$

where $L(S, t)$ is the **leverage function**.

Calibration of the Leverage Function: To match the European option prices, the local variance must be equal to the conditional expectation of the stochastic variance:

$$\sigma_{\text{loc}}^2(S, t) = L(S, t)^2 \mathbb{E}[v_t \mid S_t = S]$$

This leads to the calibration equation:

Σ Core Formula

$$L(S, t) = \frac{\sigma_{\text{loc}}(S, t)}{\sqrt{\mathbb{E}[v_t \mid S_t = S]}}$$

Numerical Calibration via Particle Method

The conditional expectation is evaluated using a Monte Carlo particle simulation: 1. Simulate N paths of (S_t, v_t) step-by-step from $t_k \rightarrow t_{k+1}$. 2. At step t_k , group the particles into bins according to their spot values $S_{t_k}^{(i)}$. 3. For each spot level S , calculate the average variance of the particles in the local neighborhood of S (using kernel density estimation):

$$\mathbb{E}[v_t \mid S_t = S] \approx \frac{\sum_{i=1}^N v_t^{(i)} K_h(S_t^{(i)} - S)}{\sum_{i=1}^N K_h(S_t^{(i)} - S)}$$

4. Evaluate $L(S, t)$ using the Dupire local vol $\sigma_{\text{loc}}(S, t)$. 5. Use the calibrated $L(S, t)$ to propagate the spot to the next step t_{k+1} .

7.4 Non-Convexity and Optimization in Calibration

Calibrating Stochastic Volatility models like Heston is highly non-convex, meaning the objective function has multiple local minima. - **Problem:** Gradient descent methods (like Levenberg-Marquardt) will get stuck in the nearest local minimum. - **Solution:** We run a **Differential Evolution** global search first.